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Dispersion performances and adsorption behavior of novel block polycarboxylate with hybrid side chains in coal-water slurry

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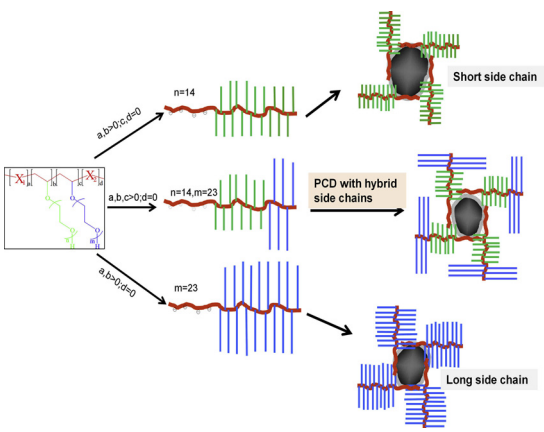
GRAPHICAL ABSTRACT



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ABSTRACT

A serial of novel block polycarboxylate dispersants (PCDs) with hybrid side chains employing long-short composite side chains for making coal-water slurry (CWS) were synthesized and applied to a low rank coal in the north of China to make slurry. The relationship of the fine structure of side chains and properties of the CWS were investigated. The results show that the apparent viscosity, zeta potential and stability of the CWS are impacted by the molar ratio of short side chain to long side chain of the PCD. Meanwhile, the viscosity reduction, stability property and zeta potentials of the CWSs are increased in accord with the order of adsorption amount of PCD on coal. These facts confirm that the PCD with a molar ratio of the short side chain to long side chain of just over unity more easily forms a larger adsorption amount on the coal surface and has optimal dispersion and stability characteristics. It is helpful to regulate the adsorption behavior of polycarboxylate dispersant to improve its dispersion property. It provides a good pathway to design and prepare a highly effective dispersant for low rank coal.

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1. Introduction

Coal-water slurry (CWS) is urgently necessary to cater for the increasing use of energy and environmental demands and also to utilize the coal resources efficiently, economically, and cleanly [1]. The viscosity, fluidity and stability of coal-water slurry have an important influence on its pipeline transportation, atomization, and combustion efficiency [2]. The dispersant plays an important role in improving the properties of CWS. The influence of the dispersant on the slurrying of coal is very important, and a suitable dispersing agent is very necessary for low rank coals to make slurries [3]. On the premise of ensuring the viscosity of the slurry, the dispersant can improve the concentration of the slurry to the maximum extent [4]. Furthermore, using a proper dispersant can increase the utilization ratio of coal [5]. In the system of coal water slurry, coal particles are dispersed by using the electrostatic repulsion and the spatial resistance of the dispersant adsorbed on their surface [6]. The dispersion properties of dispersants are closely related to the adsorption configuration and assembly morphologies in the coal water interface. The factors which affect the assembly of the dispersants in the coal water interface include the surface properties of the coal, the structural characteristics of the dispersants, and the dosage and solid content of the dispersants [7]. The factors which affect the rheological properties and application performance of the CWS are mainly the amount of the dispersant on the surface of the coal particles, the adsorption amount, the adsorption strength and the thickness of the adsorbed layer. Through the study of these factors, the molecular structure can be designed to achieve the best matching of dispersant for different kinds of coal [8].

At present, in the Chinese coal industry expanding the proportion of low-rank coal-water slurry will be an important way to develop coal-water slurry technology in China. The key problem is how to improve the performance of high concentration low-rank coal slurry [1,4]. However, because of the high oxygen content and the dense surface hydrophilic groups in lower rank coal, the dispersant does not closely adsorb on the surface of coal, which leads to the easy desorption of the dispersant when the coal water slurry is at a high temperature or is in a high shear environment [9]. This makes the viscosity of the coal water slurry increase sharply, which greatly inhibits its ability to be transported and atomized. In order to meet the requirements of industrial production, it is essential to research environmentally friendly and efficient dispersants for developing the coal-water slurry industry [10].

In recent years, a serial of new anionic, amphoteric and long-side chain polycarboxylic acid dispersants were successfully synthesized for the low rank coal-water slurry [11,13], such as Shenfu coal, Bingchang coal and Inner Mongolia coal. The effect of polycarboxylic acid structure on the slurry formation of low-rank coal was systematically studied. Furthermore, the dispersion mechanism of the polycarboxylate dispersant was discussed. The optimum ratio of ionic side groups in the polycarboxylate was found in the study of low rank coal slurry [12]. At the same time, the polycarboxylic acid with moderate side chain length had better dispersion properties, viscosity reduction and stability than others [13]. Based on the particular advantages of long and short chains, polycarboxylic acid with composite long and short side chains can play a synergistic role in dispersion for CWS. The longer side chain is beneficial to greater space resistance, but it is not conducive to the increase of the main chain polymerization degree and the decrease of the main chain. On the contrary, the shorter side chain is not conducive to the formation of stronger steric hindrance. Therefore, it is very important to combine the advantages of long chains and short chains of the polycarboxylate.

The adsorption of dispersant molecules on the surface of coal particles is a prerequisite for good dispersion properties. The properties of polycarboxylates as dispersants are largely determined by their adsorption properties at the interface of coal and water [8]. However, since there is no full understanding of the details of the adsorption process related to the action mechanism of the dispersant, precise control of adsorption behavior is often a challenging job. As the adsorption properties are dependent on the chemical structure of the dispersant [14], the influence of the composite structure of long-short chains of the polycarboxylate on its adsorption on the coal surface is well worth researching.

In this study, novel block polycarboxylate dispersant containing hybrid side chains was first synthesized, which combines the dispersing property of the long side chain and the short side chain dispersant. By analyzing the dispersing properties of the PC dispersants having various molar ratios of long side chain to short side chain, the influence of the dispersant structure on the adsorption is discussed. Meanwhile, the relationship between the adsorption amount and the viscosity, stability and the zeta potential is developed to understand the details of the adsorption process related to the action mechanism of the dispersant. It is helpful to regulate the adsorption behavior of polycarboxylate dispersant to improve its dispersion characteristic. Furthermore, it provides a good pathway to design and prepare a highly effective dispersant for low rank coal.

2. Experimental

2.1. Materials

Allyl polyoxyethylene ethers (APEG) with different molecular weights (M_w = 600,1200 were purchased from Jiangsu Haian Chemical ReagentChina; methacrylic acid MAA was purchased from Tianjin Chemical ReagentChina; Azobisisobutyl hydrazine dihydrochloride AIBA as initiator was purchased from Shanghai Macklin Biochemical Co., Ltd. S-S'-bis a,a'-dimethyl-a"-acetate- trithiocarbonate BDATT as chain transfer reagent was synthesized according to the literature [15,16].

2.2. Preparation

2.2.1. General procedure for preparation of a typical polycarboxylate dispersant

The short chain macromonomer APEG600 (n = 14) and a certain amount of water were added to a flask with magnetic agitator, and reflux condensing tube under a continuous nitrogen atmosphere. Then reaction solution in the flask was heated and stirred to dissolve it fully, then its temperature was raised. After that, a certain amount of methacrylic acid (MAA) aqueous solution, BDATT as RAFT reagent and AIBA dissolved in deionized water as the initiator were added to the mixture solution drop by drop. After the mixture solution was kept at a fixed temperature in nitrogen atmosphere, the product of step 1 was obtained. The step 1 production was purity through precipitated in ethyl alcohol and dried in vacuum. Then the long chain macromonomer APEG1200 (m = 23) and AIBA which were dissolved in deionized water were dropped into the product of step 1 then held at a constant temperature. After the pH value was adjusted to about 7–8 using 30 % NaOH aqueous solution, the hybrid polycarboxylic acid dispersants PCD containing composite long and short side chains was obtained (see Fig. 1).

In order to choose the premium ratios of short side chain to long side chain, a series of polycarboxylic acid dispersants PCD-1 ~ PCD-5 containing various ratios of short side chain to long side chain were synthesized according to the different ratios of monomers in polymerization. As controlled samples, a series of random free copolymers $PC_{Ran}1 \sim PC_{Ran}5$ were prepared with MAA, APEG600 and APEG1200 by traditional free polymerization.

2.2.2. Coal sample and coal-water slurries preparation

A low rank coal, Shenfu coal was choose to be used in this work. Raw Shenfu coal was dried in the air, and what is more it was crushed and sieved with different meshes. The results of the proximate and ultimate analyses of Shenfu coal samples were shown in Table 1. The

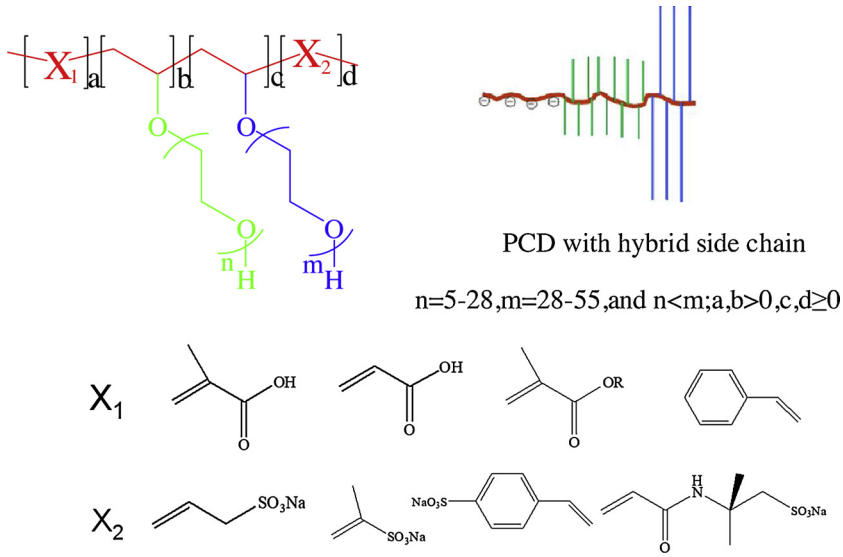

Fig. 1. The chemical structure of a typical hybrid polycarboxylate dispersant.

Table 1
Proximate and ultimate analyses of Shenfu coal sample on air dried basis.

	Item	Content
Proximate analyses	Inherent moisture (wt%)	9.63
	Ash (wt%)	14.11
	Volatile matter (wt%)	37.67
	Fixed carbon (wt%)	52.70
Ultimate analyses	Carbon (wt%)	65.12
	Sulfur (wt%)	0.41
	Hydrogen (wt%)	4.55
	Oxygen (wt%)	14.82
	Nitrogen (wt%)	0.97
	Oxygen/carbon ratio (wt%)	0.23

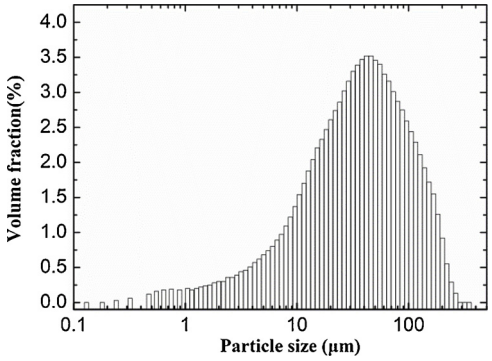

Fig. 2. The distribution of Shenfu coal particle size.

particle size distribution of Shenfu coal sample was determined via the laser particle size analyzer (BT-9300Z, Dandong, China). The result was shown in Fig. 2. The average size of volume for the coal particles was about 25.35 μm. The coal powder was slowly mixed in a pot containing a certain quantity of PCD and deionized water. The mixture was continuously stirred by a mixer during the addition of coal [17], and then the stirring of the slurry was continued for another 10 min at 1200 rpm to ensure the homogenization of the CWS. The slurry was then studied to understand its characteristics.

2.3. Measurement

2.3.1. Structure characterization of PCD

The above PCD was purified by precipitating in the solvent of dichloromethane and acetone. Then it was dried in a liquid nitrogen cold trap at a temperature of −70 ℃ and a vacuum of -0.5 MPa. The powder of the dried and pure PCD was ground with the dried potassium bromide then pressed into a sheet to test infrared spectroscopy using VECTOR-22 infrared spectrometer produced by Bruker, Germany. After the 15 − 20 mg of the powder of the PCD was completely dissolved in deuterated dimethyl sulfoxide (DMSO), it was transferred into the nuclear magnetic tube. The chemical structure was analyzed by a 400 MHz nuclear magnetic resonance instrument (NMR). The purified and dried PCD was dissolved in 0.1 mol·L^{-1} $NaNO_3$ aqueous solution. Then it was measured by gel permeation chromatography (GPC) for relative molecular weight at the test temperature of 30℃, using 0.1 mol·L^{-1} $NaNO_3$ solution with the flow rate of 0.5 mL·min^{-1} as a mobile phase. The injection quantity of the sample was 20 μL. The molecular weight was quantified by the polyethylene glycol standard samples.

2.3.2. Viscometric measurements

The apparent viscosity of coal-water slurries was tested by rheometer (the United States, Brookfield, R/S-SST Plus), at 25 ℃. The CWS was prepared at solid content of 64.0–69.0 wt% and the polycarboxylic acid dispersant (solid content calculated to be 30 wt%) was added into CWS at a dosage of 0.10-0.50 wt % to comply with the needs of the coal gasification industry. The viscosity value at a shear rate of 100s^{-1} was used as the apparent viscosity of the suspension, unless otherwise stated [13].

2.3.3. Stability measurements

The static stability of the CWS was evaluated by the water separating test (water separating ratio, v/v%) and 72 h (3 days) storage time (h) in a similar manner for all samples at room temperature [19]. The static stability of the CWS is inversely proportional to its water separating ratio. The static stability of CWS was determined by a water separating ratio W_b. The stability tends to increase with the decrease in the water separating ratio. The smaller W_b is, the better stability of the CWS. The water separating ratio was calculated according to Eq. (1).

$$W_b(\text{v/v\%}) = V/V_0 \times 100\ \% \quad (1)$$

where V (ml) was referred to the volume of upper water in the CWS

after 72 h, V_0 (ml) was the total volume of the CWS.

2.3.4. Adsorption experiments

The adsorbed amount of the PCD on coal particles was calculated by determining the amount of free polycarboxylate in the solution. The concentration of PCD in the solution was measured using total organic carbon analyzer (Elementar vario TOC cube, Germany) and calculated according to Eq. (2).

$$TOC = Kc + b \tag{2}$$

where *TOC* is the total organic carbon in the dispersant solution ($mg{\cdot}L^{-1}$) ; *K* is the factor of proportionality of *TOC* to the concentration of dispersant ; *c* is the concentration of the dispersant solution ($mg{\cdot}L^{-1}$);b is the constant of proportionality of *TOC* to the concentration of dispersant($mg{\cdot}L^{-1}$).

The fraction of organic compounds eluted from the coal in the coal water mixture without a polymer dispersant was measured as a blank value to offset background absorption of dissolved matter from coal. The amount of polymer adsorbed on coal particles was tested and calculated according to the literature [8]. Then adsorption isotherm of the polycarboxylate dispersant on coal was obtained using the fitting of the adsorption at various original concentrations of PCD.

2.3.5. ESEM–EDS measurement

Raw coal samples and complex coal samples which were adsorbing the dispersant PCD-2, were filtered from coal-water slurries and washed three times by water, then vacuum dried into dry particles. After that, the particles were pressed on conductive tapes and observed by Q45 Environment Scanning Electron Microscopy ESEM and Energy Dispersive Spectrometer EDS from FEI, US. The magnification was fixed at 2400× [18].

2.3.6. Zeta potential measurements

The zeta potential of the coal particles in the CWS was determined by ZEN3690 Zetasizer (Malvern Instruments Corp., British) [19]. Three zeta potential measurements were made, and then the mean value of them was utilized in this study.

3. Results and discussion

3.1. Characteristics of dispersants

The two types of polycarboxylic acid dispersants PCD-1 ~ PCD-5 and $PC_{Ran}1$ ~ $PC_{Ran}5$ were prepared to study the effect of the various ratio of short side chain to long side chain on PC dispersants (in Table 2). As shown in Table 2, the two types of PCs respectively are synthesized by RAFT polymerization and random polymerization. The relative molecular weight of the PC_{Ran} by random polymerization is about equal to that of the PCD by RAFT polymerization. However, the molecular weight distribution of the PCD is relatively narrower than that of the PC_{Ran}. Every molecular weight of PCD is about 10,000. Besides, each of their molecular weight distribution is close to 1.

Fig. 3a shows the infrared spectrum of PCD by the RAFT polymerization. The structure analysis results of spectra were as follows: The stretching vibration absorption peak of carboxylic hydroxyl is at 3560.90 cm^{-1}. The stretching vibration absorption peaks of the carbon-hydrogen bond and carbon-oxygen bond in ether are respectively at 2863.12 cm^{-1} and 1103.65 cm^{-1}. The carbonyl group stretching vibration peak is at 1728.04 cm^{-1}. No peak of carbon-carbon double bond that proves the reaction of all monomers is completed. The methylene carbon-hydrogen bending vibration absorption peak is at 1470.44 cm^{-1}. The carbon-hydrogen bending vibration absorption peak of methyl is at 1351.98 cm^{-1}.

As presented in Fig. 3b, the chemical shift δ is the hydrogen proton peak in the polymer between 0.9–1.1 ppm. The peak between 1.6–1.9 ppm is the hydrogen proton of the main chain of the PCD. The peaks of 2.0–2.2 ppm are attributed to the hydrogen proton of methine between the polyether side chain and the polymer main chain. The peaks of 3.5–3.7 ppm are the hydrogen proton of hydroxyl at the end of the polyether side chain. The peaks of 3.8–4.0 ppm are the hydrogen proton of methylene close to the hydroxyl in the polyether chain. The peaks of 4.5–4.8 ppm are the hydrogen proton of methylene which connected with the ether bond in the polyether chain. No peak is found at about 5.7 ppm (i.e. the hydrogen related to the double bond), which represents the disappearance of double bonds. Based on the analyses of ^{1}H-NMR and FT-IR spectra, it is clearly seen that the synthesized polymer is the target polycarboxylic acid.

Table 2
Molecular weight and distribution results for two types of polycarboxylic acid dispersants.

Entry	n_{MAA} :$n_{APEG600}$:$n_{APEG1200}$	PC_{Ran}			PCD		
		M_w	M_n	PDI	M_w	M_n	PDI
#1	4 :1: 0	10154	6550	1.55	9854	9040	1.09
#2	4 :0.75:0.25	12030	7613	1.58	11667	10324	1.13
#3	4 :0.5: 0.5	13357	8348	1.60	12507	10336	1.21
#4	4 :0.25 :0.75	13117	8408	1.56	12300	9840	1.25
#5	4 :0:1	14322	9007	1.59	13117	10168	1.29

3.2. Slurry properties of CWS with PCD

Fig. 4a indicates the effect of the polycarboxylic acid dispersants prepared by two methods on apparent viscosity of the CWS. The average value of the apparent viscosity of the CWS was obtained when the slurry concentration was 66 wt% and the dispersant content was 0.4 wt%. As Fig. 4a shows, the effects of the dispersants synthesized by the two methods on the concentration of coal slurry are definitely different. The apparent viscosity of the CWS with the PCD synthesized by the RAFT method is lower than that of PC_{Ran} synthesized by random copolymerization under the same molar ratio of long and short side chains. The results show that the PCD has a better viscosity reduction effect on coal-water slurry than the PC_{Ran}. The stability results of the above CWSs in Fig. 4b were characterized by the water separating ratio method. Under the same molar ratio of long and short side chains, the water separating ratio of the CWS with the PCD is less than that of $PC_{Ran.}$ It means that the stability of the CWS with the PCD is better than that of $PC_{Ran.}$ By comparing the apparent viscosity and dispersion stability of CWSs using PCD with different molar ratios of short and long side chains, it can be concluded that the PCD-2, PCD-3 and PCD-4 with the hybrid-length side chains has better dispersion stability than that of other the polycarboxylic acid with only long side chains (PCD-5) or only short side chains (PCD-1). Especially, PCD-2 shows the best viscosity reduction effect and dispersion stability which may be due to the absorption behavior of PCD on coal particles surface.

As shown in Fig. 4c, the apparent viscosity of the CWS in presence of PCDs having composite side chains decreased with increasing shear rate. The CWS behaves as a non-Newtonian fluid with shear thinning. Two most widely used rheological models, Herschel-Bulkley and Power-law used in CWS, were applied to fit the curves of stress-shear rate in Fig. 4d. The fitting results were shown in Table 3. As presented in Table 3, the rheological properties of the coal water slurry are more consistent with Herschel-Bulkley because the correlation coefficient (R^2) of the Herschel-Bulkley model was closer to 1 than that of Power-law model. Meanwhile, as the flow behavior index (n) was less than 1, the coal water slurry behaved as a pseudoplastic fluid which is good for stability and storage of CWS. Besides, as shown in Table 3, the order of apparent viscosity of CWSs at shear rate of $100s^{-1}$ from lowest to highest is PCD-2 < PCD-3 < PCD-4 < PCD-1 < PCD-5, which is in agreement with the consistency coefficient (*K*). According to reference [20], the CWS with the larger flow characteristic index has better

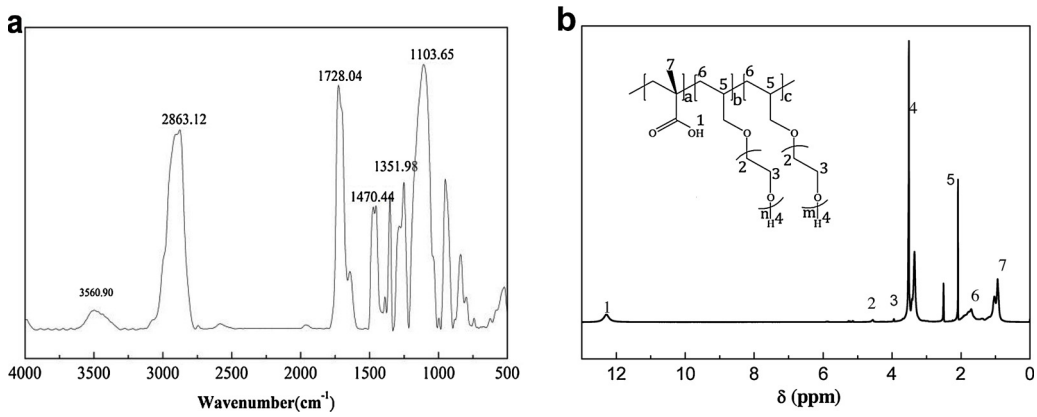

Fig. 3. The FT-IR spectra **(a)** of PCD-2 and **(b)**^{1}H-NMR of PCD-2.

flowability. Thus we can conclude that the CWS with dispersant PCD-2 has better flowability because it has a larger flow behavior index. The reason for this may be diverse adsorption of PCD on the coal surface [21].

3.3. Adsorption studies

3.3.1. Adsorption morphology

The ESEM images and EDS analysis of raw Shenfu coal powder and complex coal powder which adsorbed the PCDs are presented in Fig. 5. It can be seen from Fig. 5a that there are no gaps between the raw coal particles which aggregate together. When the dispersant PCD-2 was adsorbed on the coal to form complex coal (Fig. 5b), the intergranular gaps between the complex coals are observed. As a result, the dispersion of coal powder becomes better. It may be because the large specific surface of the coal powder is reduced after the surface of coal is modified by the CD. Therefore, the PCD weakens the aggregation of coal particles and disperses them in coal water slurry [22]. Based on the element content on the coal particle surface from EDS analysis (Fig. 5c), there are a fraction of ionic groups on coal surface containing oxygen such as carboxyl group, hydroxyl group and salts. Furthermore, it can be seen that the elements on the surface of the complex coal particle

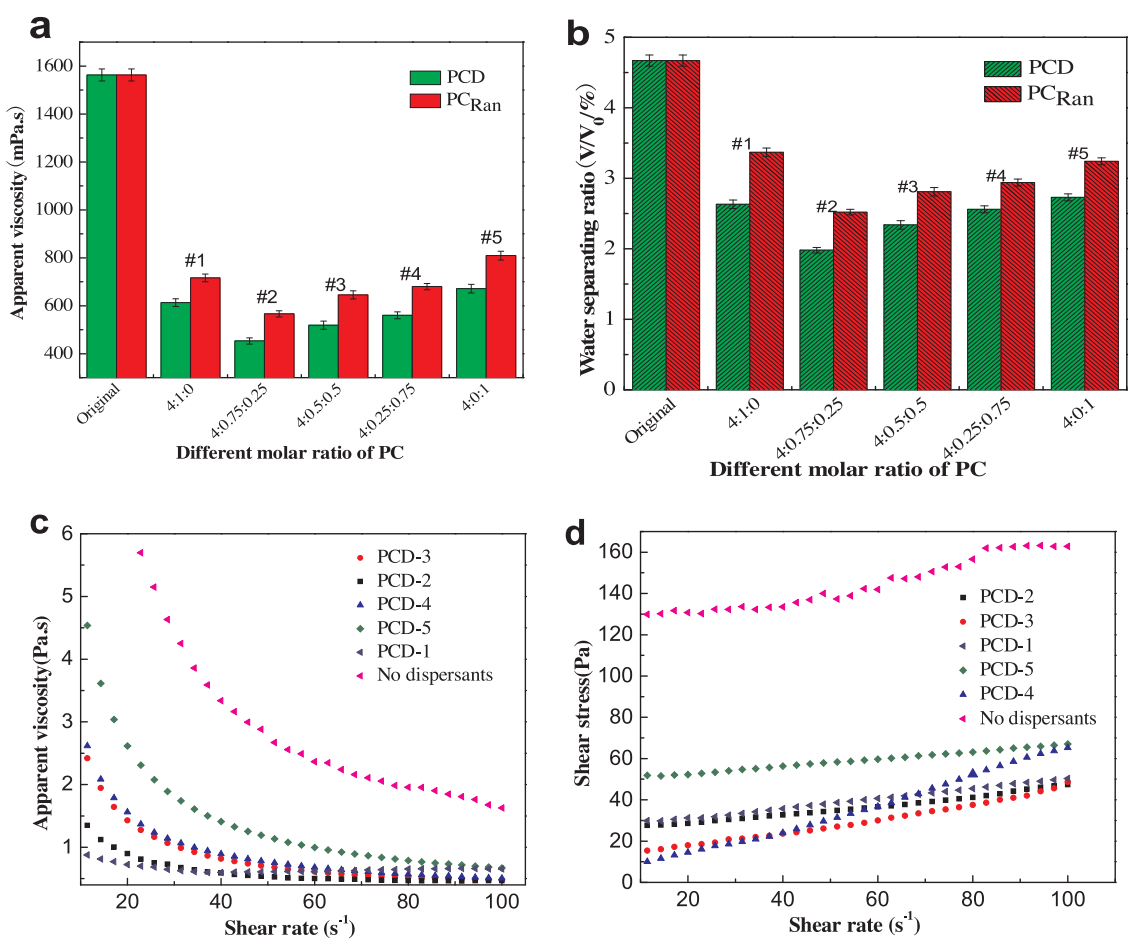

Fig. 4. Apparent viscosity **(a)** and stability **(b)** of CWSs with 0.4 wt% PCD and PC_{Ran} synthesized by random polymerization under the conditions of 66 wt% coal concentration, pH 7, 25 °C, and shear rate of 100s^{-1}. Apparent viscosity **(c)** and shear stress**(d)** as functions of shear rate for coal slurries with 0.4 wt% PCDs having composite side chains under the conditions of 66 wt% coal concentration, pH 7, 25 °C, and shear rate from 0 to 100s^{-1}.

Table 3
Rheological properties of coal-water slurry fitting results.

Dispersant	Apparent viscosity	Herschel-Bulkley				Power-law		
	(mPa·s)	τ_0/Pa	K/Pa.s^n	n	R^2	K/Pa.s^n	n	R^2
PCD-1	613	27.08	0.1552	0.8796	0.9977	16.07	0.2330	0.8927
PCD-2	471	26.28	0.0429	0.8324	0.9969	15.08	0.2272	0.8529
PCD-3	519	14.38	0.0496	0.8665	0.9973	3.35	0.5522	0.9194
PCD-4	560	6.03	0.1214	0.8741	0.9985	0.59	1.0191	0.9909
PCD-5	671	52.06	0.1853	0.9299	0.9841	42.3	0.0866	0.7043

(Fig. 5d) change obviously compared with that of raw coal particles (Fig. 5c). As presented in the tables insert in Fig. 5c and d, the amount of C*K* and S*K* increase obviously, however, the amount of Al*K* and Si*K* decrease. This is due to the fact that the relative contents of carbon and sulfur is higher than that of raw coal. Furthermore, there is no Al*K* and Si*K* in the PCD-2 molecule. It is further confirmation that coal powder is dispersed because the PCD has been adsorbed and assembled on the surface of the coal particles.

3.3.2. Adsorption amount

The isothermal adsorption of five kinds of PCD dispersants containing composite long and short side chains on Shenfu coal at 25 °C is shown in Fig. 6a. The adsorbed amount of all the five kinds of PCDs rapidly increases as the PCD concentration increases from 0 to 600 mg L^{-1}. The appearance of plateau adsorption in the adsorption isotherm for concentrations larger than 600 mg L^{-1} suggests the adsorption curves follow a monolayer adsorption model. The results are consistent with the results which were obtained in the reference [21]. The adsorption of dispersant on coal conforms to the *Langmuir* adsorption model (as shown in Eq. (3), so the isothermal adsorption of five different dispersants was fitted according to *Langmuir* isothermal adsorption model (Fig. 6a). The fitting results of isothermal adsorption of the PCDs are shown in Table 4. The *Langmuir* isothermal adsorption formula is

$$\frac{1}{\Gamma} = \frac{1}{\Gamma_\infty} + \frac{1}{\Gamma_\infty Kc} \tag{3}$$

Where, Γ_∞ is the statured adsorption amount, mg·g^{-1} ; Γ is the equilibrium adsorption amount, mg·g^{-1} ; c is the equilibrium concentration of PCD, mg·L^{-1} ; K is the equilibrium adsorption constant, L·g^{-1}。

The *Langmuir* model is very suitable for fitting adsorption isotherms of the PCDs on the Shenfu coal as all the *Langmuir* fitting correlation coefficients (R^2) are quite close to 1. Furthermore, the curves of $1/\Gamma \sim 1/c$ were drawn according to Eq. (2). The linear fitting curves of $1/\Gamma \sim 1/c$ are shown in Fig. 6b. The saturated adsorption amount and equilibrium constant of dispersant on coal were calculated and the results are shown in Table 4. The saturated adsorption amount of the five PCDs on Shenfu coal decreases in the order of PCD-2 > PCD-3 > PCD-4 > PCD-1 > PCD-5. So the order of the saturated adsorption amount corresponds well with the order of the apparent viscosity. The saturated

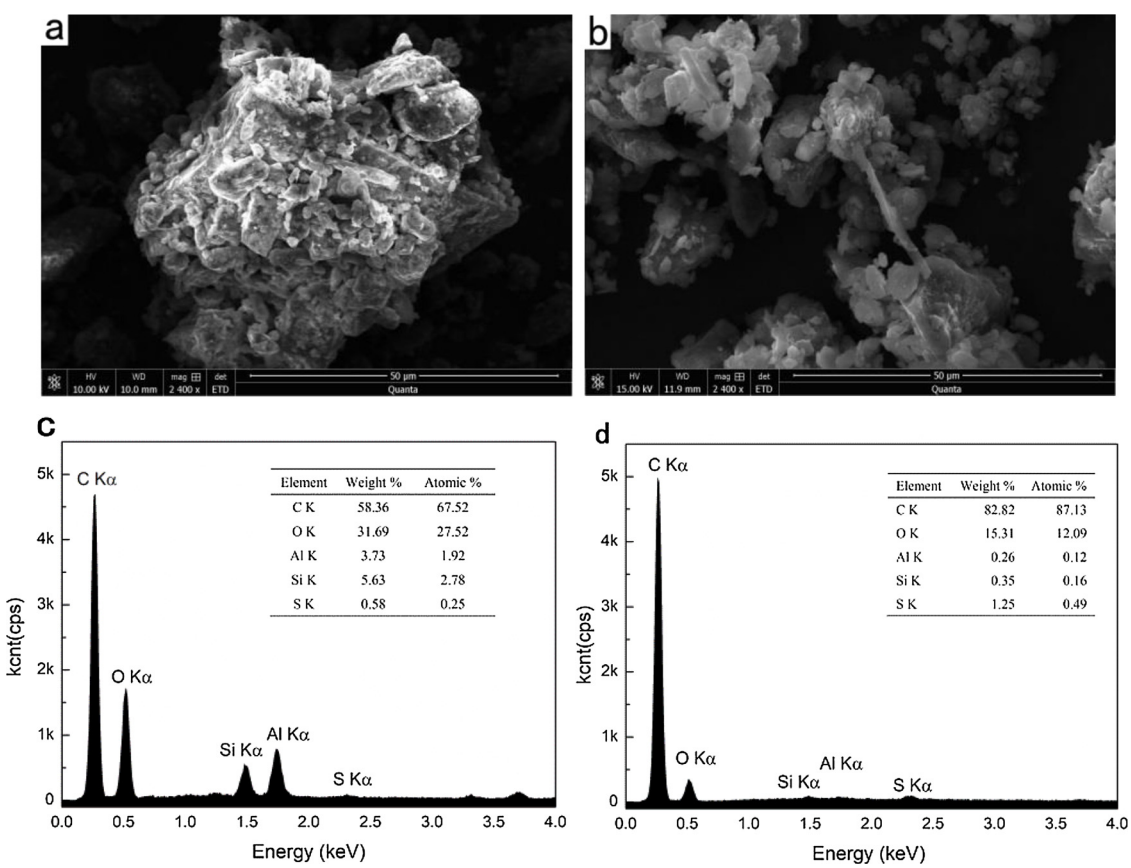

Fig. 5. ESEM-EDS of raw coal and coal with PCD adsorption;**(a)**ESEM image of raw Shenfu coal; **(b)**ESEM image of Shenfu coal adsorbing PCD;**(c)**EDS image of raw Shenfu coal;**(d)**EDS image of Shenfu coal adsorbing PCD.

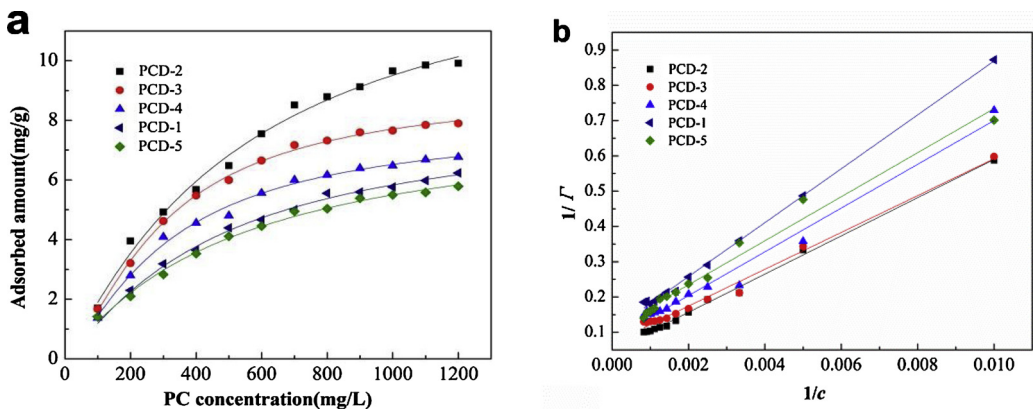

Fig. 6. (a) Adsorption of PCDs having composite side chains on coal particles, which was studied under the conditions of pH 7, 25 °C and a coal concentration of 66 wt%. **(b)** The linear fitting curves of $1/\Gamma \sim 1/c$ under the conditions of pH 7, 25 °C and a coal concentration of 66 wt%.

Table 4
Linear fitting results of isothermal adsorption equations of PCDs.

PCD@coal[a]	Γ_∞/mg·g^{-1}	K/L g^{-1}	R^2
PCD-1@coal	9.48	0.138×10^{-2}	0.99811
PCD-2@coal	20.64	0.894×10^{-3}	0.99628
PCD-3@coal	14.30	0.133×10^{-2}	0.99038
PCD-4@coal	12.49	0.129×10^{-2}	0.97878
PCD-5@coal	9.10	0.176×10^{-2}	0.97671

[a] PCD@coal means the PCD adsorbs on coal surface.

adsorption amount of the PCD-2 on the Shenfu coal is 20.6398 mg g^{-1} which is larger than others. The saturated adsorption amount decreases in turn as the molar ratio of the long chain increases. It may because the short side chain helps enhance the electrostatic interaction between PCD molecules and the ionic structure of the minerals on the coal surface [23]. As mentioned in the literature [13], as the molar ratio of the long chain increases, the carboxyl groups of the PCD are wrapped by the long chain. Consequently, the electrostatic attraction of the carboxyl groups with coal particle will be weakened.

3.4. Effect of adsorbed amount on apparent viscosity of CWS

The apparent viscosity of the CWS and the adsorbed amount of PCDs corresponding initial concentrations of PCDs including 0, 200,400,600,800 and 1000 mg/L were used to exploit effect of adsorbed amount on apparent viscosity of CWS. Fig. 7 shows that influence of adsorbed amount of PCDs on the apparent viscosity of the CWS. The apparent viscosity of the CWS without any PCD was up to 1560 mPa·s. The adsorbed amount of PCD increases with the initial concentration of PCD increases from 200 to 1000 mg/L. As a result, the apparent viscosity of the CWS also dropped with the increase of PCD adsorption. It depends on the effective utilization of PCD. In the same initial concentration of PCD, the higher adsorption amount indicates the higher utilization. The order of the adsorption amount corresponds well with the order of the apparent viscosity. The addition of PCDs reduced the attraction forces between the coal particles by generating a steric effect via adsorbing on their surface (the adsorption amount in Fig. 6 and Table 4). The more significant viscosity change is also attributed to the higher adsorption of PCDs on coal. The order of dispersants in terms of increasing viscosity is PCD-2 < PCD-3 < PCD-4 < PCD-1 < PCD-5. This is in accord with the order of adsorption amount of PCD on coal in each same initial concentration of PCD. Therefore, the highest adsorption amount of PCD-2 achieves its highest utilization.

3.5. Effect of adsorbed amount on stability of CWS

The influence of adsorbed amounted of PCDs on the stability of CWS is shown in Fig. 8. In each same initial concentration of PCD, the water separating ratio of CWS is decreasing with the increasing of the adsorbed amount of PCDs on coal. Because the static stability of the CWS is inversely proportional to its water separating ratio, the order of stability is PCD-2 > PCD-3 > PCD-4 > PCD-1 > PCD-5 which is in

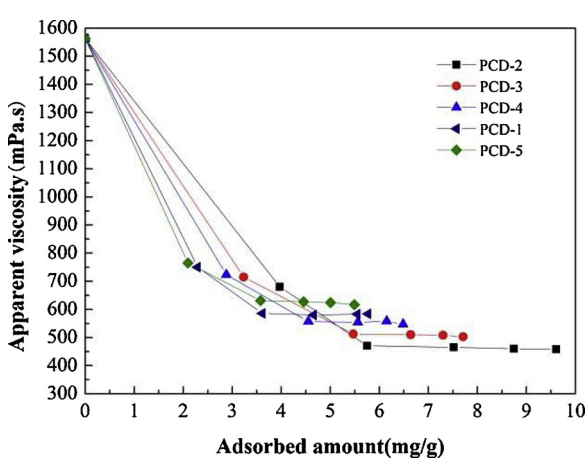

Fig. 7. Influence of adsorbed amount of PCDs on the apparent viscosity of coal having different particle sizes with pH 7, 25 °C, 100 1/s shear rate and a 66 wt% coal concentration.

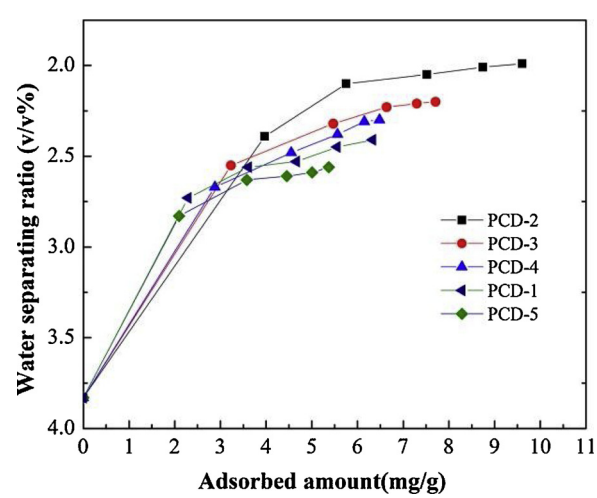

Fig. 8. Effect of adsorbed amount of PCDs on water separating ratio of CWSs with pH 7, 25 °C and a coal concentration of 66 wt%.

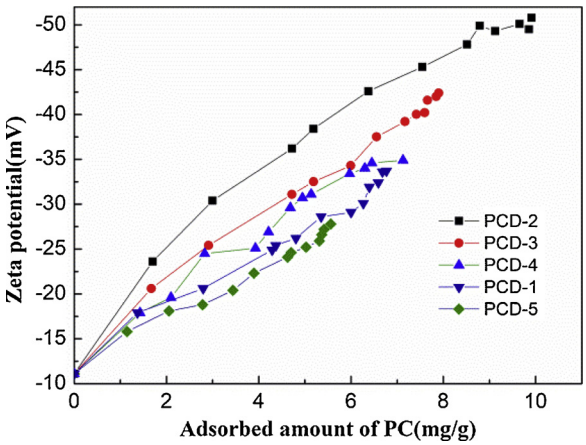

Fig. 9. Effect of adsorbed capacity of PCD on the zeta potential of the coal particles which was studied under the conditions of pH 7, 25 °C and a coal concentration of 66 wt%.

accord with the order of adsorption amount of PCD on coal. These results confirm that this increase in stability of coal particles is due to steric effect generated between the particles by the adsorbed PCDs with composite long-short side chains [24].

3.6. Effect of adsorbed amount on zeta potential analysis

Fig. 9 presents the effect of the amount of dispersant adsorption on zeta potential. Shenfu coal as a low rank coal was chosen to be used in this work. The value of zeta of the CWS present negative (−11 mV) without any dispersants based on SEM/EDS results of ionic groups on coal surface containing oxygen such as carboxyl group, hydroxyl group and salts. With the increase of adsorption amount, the absolute value of zeta potential on the coal surface increases. Therefore, the adsorption amount can promote the zeta potential of the coal surface to some extent. This is because with the increase of the amount of dispersant adsorbed on the coal surface, the interaction between the dispersant and the coal particle surface increases, and the amount of dispersant adsorbed per unit area increases [25]. In this way, the negative charge on the dispersant increases per unit area, and the absolute value of zeta potential on the coal particle surface is increased. The electrostatic repulsion between dispersants is enhanced with the increase of short side chains. The order of the zeta potential absolute value, PCD-2 > PCD-3 > PCD-4 > PCD-1 > PCD-5, is consistent with the order of adsorption amount of PCD on coal. As the molar ratio of the long chain APEG1200 increases from 0 to 1, the electrostatic attraction of the carboxyl groups of the PCD with coal particle will be weakened. This result is caused by carboxyl groups being wrapped by the long chains [26]. It is thus apparent that the short side chain helps enhance the electrostatic interaction between PCD molecules and the ionic structure of the minerals on the coal surface.

3.7. Mechanism of dispersion

These results indicate that the adsorption of PCDs affected both the steric hindrance of the particles and the electrostatic repulsion between particles via adsorption of dispersant on the surface of the coal particles [27]. The adsorption affected the steric hindrance more significantly than the electrostatic repulsion [28]. The longer side chains in the block PCDs provide a larger space steric hindrance, which gives the coal-water slurry good dispersion. The existence of shorter side chains improved the electrostatic repulsion between dispersants and the dispersion stability of coal-water slurry [29]. It can be proved by the adsorption amount of dispersant that the dispersant with composite long and short side chains has higher adsorption amount, so that there are more dispersants attached per unit area to enhance the dispersion of the coal-water slurry system [30]. The block PCDs with composite long-short side chains combine the advantages of the long and short side chains. The PCD-2 with desirable proportion of shorter side chain to longer side chain shows the best viscosity reduction effect and dispersion stability. The shorter side chains of PCDs towards to coal can easily enter the coal surface, thus greatly increasing the adsorption amount, but the longer side chains back to coal can provide a certain steric resistance and increase the repulsive effect between coal particles. The stability of the system is increased in accord with the order of adsorption amount of PCD on coal.

4. Conclusions

This study has shown that novel hybrid polycarboxylate dispersant containing composite long-short side chains was successfully synthesized by RAFT polymerization. The rheological properties of the coal-water slurry with novel hybrid are more consistent with Herschel-Bulkley. The CWS with behaves as a pseudoplastic fluid which is good for storage and pumping. The research has also shown that the viscosity reduction and stability property of the CWS is increased in accord with the order of adsorption amount of PCD on coal. The adsorption of PCDs affected the steric hindrance and wettability of the particles as well as the electrostatic repulsion between particles via adsorption and assemblage of dispersant on the surface coal particles.

The evidence from this study suggests that novel block PCD combines both the properties of the block of long side chain and the block of short side chain. In particular, the PCD with slightly more short side chains than long side chains more easily forms a thick adsorption layer on the coal surface. It is helpful to regulate the adsorption behavior of polycarboxylate dispersant to improve its dispersion property. The findings of this research provide a good pathway to design and prepare a highly effective dispersant for CWS industry.

CRediT authorship contribution statement

Junfeng Zhu: Conceptualization, Project administration, Writing - review & editing. **Weichun Gao:** Formal analysis, Visualization. **Pei Wang:** Writing - original draft, Methodology. **Ting Zhu:** Formal analysis. **Junguo Li:** Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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